

Semester Paper (2023)

Vertex-critical graphs that are not edge-critical

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Date:	August 31, 2023

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Abstract

This paper is a summary of a problem first proposed by Dirac on whether vertex-critical graphs that were not edge-critical could exist, and how many extra edges they could have. Several results have been published concerning this problem, and we outline the results and their proofs, whilst providing diagrams to visually comprehend the constructions.

Acknowledgements. Thank you to Raphael Steiner for agreeing to supervise my semester paper and introducing me to this beautiful problem that I (unsuccessfully) tried to work on and enjoyed thinking about immensely.

Introduction

[...]Quaecunque sit fluuii figura et distributio in ramos, atque quicunque fuerit numerus pontium, inuenire, vtrum per singulos pontes semel tantum transiri queat, an vero secus?

Leonhard Euler, *Solutio problematis ad geometriam situs pertinentis*.

Gabriel Andrew Dirac was one of the foremost mathematicians in graph theory and was one of the researchers responsible for much of the early discoveries and understanding of the topic. Born Balázs Gábor in Budapest in 1925, he changed his name when his mother remarried the famous English¹ physicist Paul Dirac². He died at the age of just 59 in 1984, prompting his longtime collaborator, and the most prolific mathematician of our age, Paul Erdős, to write a short paper[5] summarising some of their shared work in tribute. Dirac and Erdős were both particularly interested in vertex-critical graphs, or graphs where reducing the number of vertices by one reduces the chromatic number of the graph by one, irrespective of which vertex is removed. Consider for example the following beautiful lemma of Dirac[4] whose proof relies on vertex-critical graphs and their properties.

Lemma 0.1. *A finite k -chromatic graph admits a graph of minimum degree $(k - 1)$ as a subgraph, and by extension all trees on k vertices.*

Proof. Every k -chromatic graph admits a k -chromatic vertex-critical subgraph. To see this, just remove vertices one by one. If the chromatic value decreases to $(k - 1)$; rewind a step and remove another vertex. Terminate when this is not possible (i.e., removing any vertex would reduce the chromatic number by one). This must happen at some point, as if the order is smaller than k then it is at most $(k - 1)$ chromatic.

If this vertex-critical subgraph has a vertex v with degree less than $(k - 1)$, we can remove it and colour the remaining vertices with $(k - 1)$ colours; v can then be coloured in one of these as well since it is only restricted from at most $k - 2$ colours, a contradiction.

This vertex-critical subgraph contains every tree on k vertices. This can be seen by induction: the case $k = 2$ is trivial. Now, for a tree of arbitrary size n , remove a leaf v previously attached to some u : by induction, this subtree is contained in every graph of minimum degree $k - 2$, including our graph with minimum degree $(k - 1)$. If we now find the subtree, then the vertex corresponding to u is connected to at least one vertex not already in the tree, which we can take as v . $\square$

Dirac conjectured, as reported by Erdős in his paper, that for every $k > 4$, one could find a k -chromatic vertex-critical graph that remained critical if any of its edges were removed. Ruminating on the matter, Erdős agreed that the conjecture was likely to hold and that if it was true, one could likely find constructions where one could remove more than one edge. He conjectured that the maximal number of edges one could always remove, irrespective of which edges were picked, was an increasing function of k .

¹Born a Swiss citizen!

²Remarkably, both his stepfather and maternal uncle would win the Nobel Prize for Physics.

1 Definitions and framework

Give me six hours to chop a tree, and I will spend the first four writing my $\text{\LaTeX}$ style file.

Provenance unknown.

For the entirety of this paper, we will assume all graphs are simple.

We include the basic definition of the chromatic number of a graph for completeness.

Definition: Chromatic number.

Given a graph $G = (V, E)$, we say the *chromatic number* $\chi(G)$ is the smallest number k such that there exists a function $f: V \rightarrow [k]$ with the property that

$$\forall (u, v) \in E, f(u) \neq f(v).$$

Intuitively speaking, the chromatic number is the least number of colours one needs to colour all the vertices such that no edge is between two vertices of the same colour.

As mentioned before, we say the graph is *vertex-critical* if removing any vertex reduces the chromatic number. For simplicity of notation, we will say k -chromatic vertex-critical graphs are *k -critical* henceforth. We will call a valid colouring of the vertices in k colours a *good k -colouring*. A *critical edge* is an edge which, if removed, reduced the chromatic number. Note that removing a single vertex or edge reduces the chromatic number by at most 1. This is easy to see: if the new graph is $(k - 2)$ -chromatic then take a good colouring and simply colour the removed vertex (or either vertex adjacent to the removed edge, if an edge was removed) in a new colour to obtain a good $(k - 1)$ -colouring.

Finally, let us state properly the two conjectures of Dirac and Erdős.

Conjecture of Dirac.

For each $k \geq 4$, there is a k -critical graph with no critical edges.

Conjecture of Erdős.

For $k \geq 4$, there is a sequence of integers $f_4, f_5, \dots$ and of graphs $G_4, G_5, \dots$ such that $\lim_{i \rightarrow \infty} f_i = \infty$ and that for any set $A \subseteq E(G_k)$, $|A| \leq f_k$ of edges, $G_k - A$ is k -chromatic.

2 Dirac's conjecture

People almost invariably arrive at their beliefs not on the basis of proof but on the basis of what they find attractive.

Blaise Pascal, *De l'art de persuader*.

Dirac's conjecture embodies a simple but compelling truth in graph theory: that vertices have a greater influence on the properties of a graph than edges do. It is trivial that if a graph has critical edges then it must also have critical vertices - just take vertices adjacent to the edges in question. Why, then, is it so hard to find a graph with only critical vertices and no critical edges?

To understand this, we must first think about the problem itself and the implications of such a construction. If all vertices are critical, the simplest consequence is that if any vertex v is removed, the remaining vertices can be coloured in $k - 1$ colours. In particular, for all such colourings of the remaining vertices, there must be an edge between v and at least two representatives of each colour.

There are two major difficulties to overcome here: ensuring $G - v$ is always colourable with less colours and ensuring no edge adjacent to v is the only edge going to some colour in a good $(k - 1)$ -colouring of $G - v$.

2.1 Brown colours: the first case solved

Brown[2] was the first to publish a construction; he noted that an earlier paper by Chvátal, Graham, Perold and Whitesides[3] on Berge's strong perfect graph theorem provided in particular an interesting way to obtain a graph construction that happened to be 5-critical but with no critical edges. Chvátal and his coauthors proposed the following definition. As is standard, let $\alpha(G)$ denote the size of the largest independent set of G and $\omega(G)$ denote the size of the largest clique in G .

Definition: Minimal imperfect graphs I.

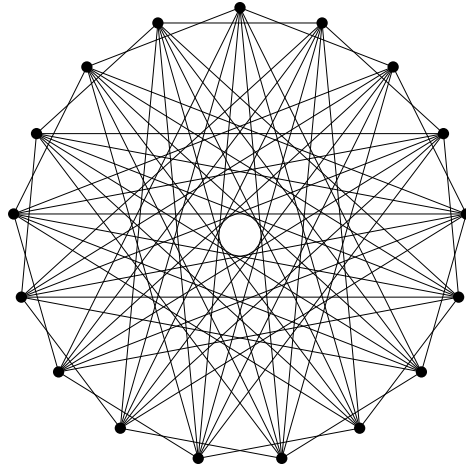
Given a graph $G = (V, E)$, we say the graph is an (x, y) -graph if it satisfies the following conditions:

- $\alpha(G) = x, \omega(G) = y$;
- $|V| = xy + 1$;
- Every vertex $v \in V$ is in x independent sets of size x and y cliques of size y ;
- There is some matching of the $|V|$ independent sets of size x and $|V|$ cliques of size y such that an independent set and a clique are disjoint if and only if they are matched.

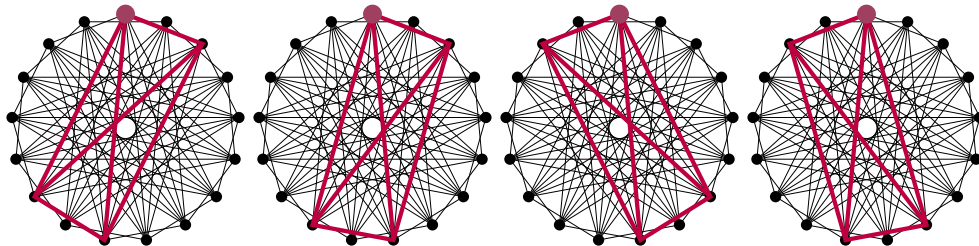
Note that the number of cliques/independent sets of largest size being $|V|$ follows from a simple double counting argument. Note also that the maximal intersection of an independent set and clique is 1 for obvious reasons. The set of all such graphs for all combinations $(x, y) \in \mathbb{N}^2$ is called the set of *minimal imperfect graphs*, and are of particular interest in the study of perfect graphs.

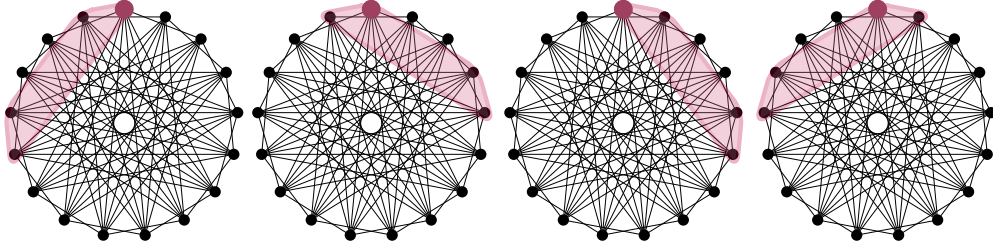
Why are these graphs desirable? Tucker showed in a later paper that for a (x, y) -graph G , and for any vertex $v \in V(G)$, $G - v$ had a unique good y -coloring. This is extremely useful - note that firstly G must be $(y + 1)$ -chromatic, by the condition on its order. Furthermore, Tucker's result shows that it is $(y + 1)$ -critical; if now every vertex v is connected to at least two vertices of every colour in the unique y -colouring of $G - v$ then it is easy to see that removing no edge can make G y -chromatic.

Consider now the following graph.

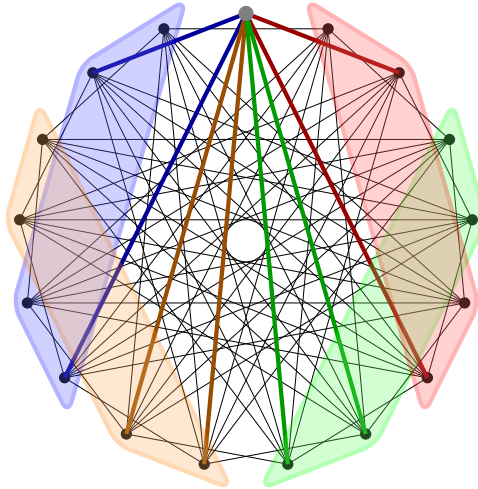


This graph G is just the Cayley graph on $\mathbb{Z}/17\mathbb{Z}$ with generating set $\{\pm 2, \pm 6, \pm 7, \pm 8\}$. In particular, since it is transitive, we know that removing any vertex has the same effect, simplifying our calculations somewhat. Let us first check it satisfies the conditions to be a $(4, 4)$ -graph. Clearly, one can see that the order is correct, and that the largest clique is of size 4 and the largest independent set is of size 4 as well. Now let us check how many cliques and independent sets each vertex belongs to.





Moreover, it is easy to see that via the structure of both the clique and the independent set, there is a one-to-one pairing between them that maintains disjointness. One can consider the independent set to be of the form $(a, a + 1, b, b + 1)$ with $b := a + 2$, giving a gap of size 2 between the two pairs. This gap cannot pass over the short edges of the clique (those of distance 2) and so the entire independent set must be on the side of the largest chord of the clique that contains no other edges from the clique. Removing a vertex, we see that there is indeed a unique 4-good colouring.



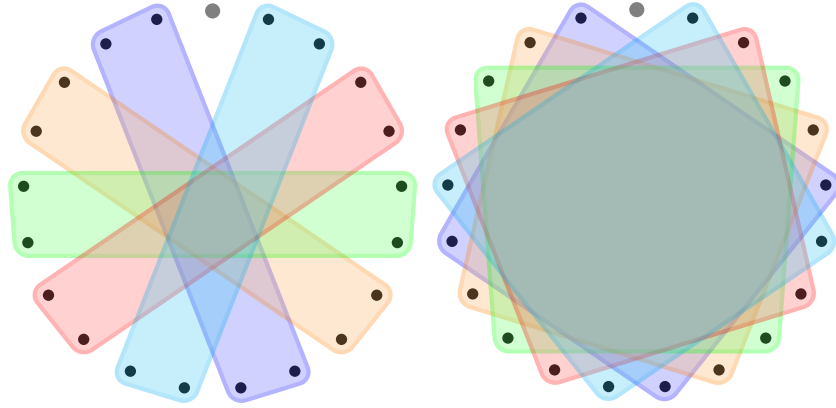
We easily see that the removed vertex is adjacent to two representatives from every colour class, proving the graph must be 5-critical with no critical edges, and thus settling Dirac's conjecture for $k = 5$.

Why can't we use this method to find constructions for larger k ? Jensen[7] actually managed to do so, finding constructions for $k = 6$ and $k = 8$. Unfortunately, he did not give the explicit construction, but just mentioned that he had a 10-regular example on 21 vertices in the first case and a 14-regular example on 29 vertices in the second. However, with this information, let us try and reconstruct the graphs, assuming they are also Cayley graphs on Z/nZ .

For $k = 6$, we know we are looking for a $(4, 5)$ -graph, such that each vertex appears in 4 independent sets of size 4 and 5 cliques of size 5. In particular, since we are working

with a symmetric transitive graph, any shift or reflection of an independent set must be an independent set. We would therefore ideally want only one configuration for an independent set, having an axis of symmetry (like in the previous case) so that exactly 4 translations of it are possible. We also want to be able to cover every vertex but one with 5 copies of this independent set. We also want our clique of size 5 to be unique up to translation, and to be symmetric under reflection and have one axis of symmetry so that only 5 translations are possible.

The easiest place to start is by testing valid configurations for the independent set. Under the constraint that it must be symmetric and provide a cover $G - 0$, we find a few possibilities. Note that henceforth when it is clear we are referring to Cayley graphs on cyclic groups, we will write $G - 0$ rather than $G - v$, and refer to the other vertices as integers as well.



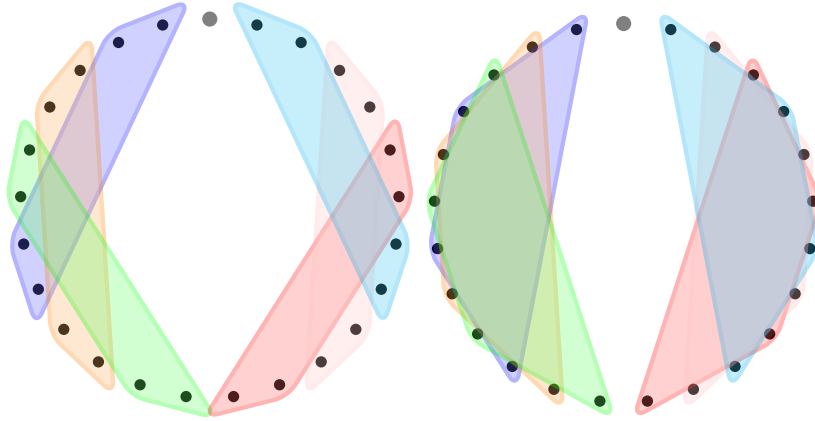
Since x is now odd, we are severely limited for choices of symmetric sets of size 4 that tile all but one vertex. Unfortunately, these two possibilities suffer from a fatal flaw that kills all “moduliesque” constructions, which is that the set of plausible edges (removing the distances inside independent sets) will never allow the removed vertex to connect twice to all the independent sets. In particular, the two adjacent sets to the removed vertex - those in blue and cyan in the above diagrams - have three vertices which will always be unattainable. This is particularly sad since we can clearly reach a $(4, 5)$ -graph with the right-hand configuration, for example (it suffices to introduce an edge between i, j iff $|i - j| < 5$). In particular, this shows that not every (x, y) -graph necessarily allows us to introduce enough additional edges.

Transitivity, therefore, is not compatible with our aims. We must therefore assume Jensen found a $(4, 5)$ -graph in another way (in particular, Chvátal et al. proposed alternative ways for finding (x, y) -graphs via linear algebra). However, just because we were defeated by the even case, we should not ignore the odd case. Notably, it seems like the case for the $(4, 4)$ -graph should be easily generalizable. Taking $k = 7$, let us begin again with the 6-colorings of $\mathbb{Z}/25\mathbb{Z} - 0$ - but this time, we do not consider the modulo-based constructions we were forced into in the odd case, and opt for symmetric constructions instead, like in Brown’s example.

2.2 Lattanzio's construction for nonprimes plus 1

Here we discuss how Brown's constructions might be generalised. In particular, Lattanzio[8] was the first to adapt Brown's construction; his construction is a very natural extension from $k = 5$ to all nonprimes plus 1.

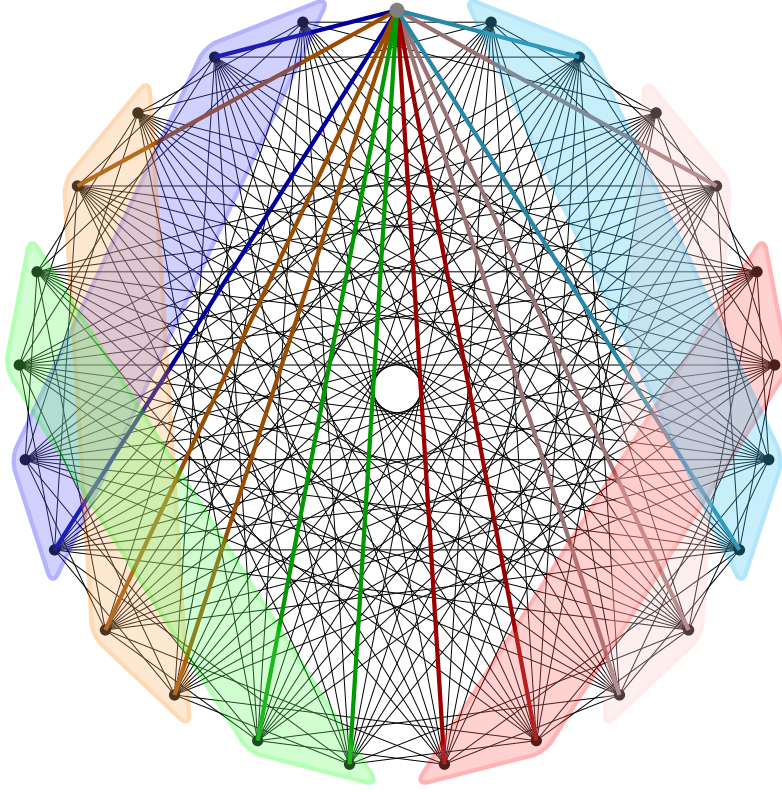
Consider the case $k = 7$ and look at the choices of a symmetric independent set that can cover $G - 0$, whilst maintaining a break down the middle as in Brown's construction.



In the first case, the internal distances within in each independent set are $\{\pm 1, \pm 5, \pm 6, \pm 7\}$, meaning the other 8 options are open to us. We must use ± 2 and ± 8 as we need v to be connected twice to the blue and cyan groups. We also want a symmetric clique of size 6, meaning a good choice would be to also use $\pm 4, \pm 9, \pm 10, \pm 11, \pm 12$ so that the only valid cliques of size 6 are of the form $\{a, a + 2, a + 4, a + 12, a + 14, a + 16\}$ and due to their reflective symmetry (across $a + 8$) it is clear that every vertex is in 6 cliques of size 6. Finally, like before, it is clear that a disjoint independent set cannot cross over the edge $(a + 4, a + 13)$ and must find itself entirely on one side of this edge, meaning there is only one independent set disjoint with the clique - $\{a + 5, a + 6, a + 11, a + 12\}$. In particular, this is a $(4, 6)$ -graph and we see that furthermore the missing vertex is connected to at least two representatives from each colour class (namely, three from the orange and pink classes).

In the second case, the internal distances within in each independent set are $\{\pm 3, \pm 6, \pm 9\}$, meaning the other 9 options are open to us. However, this immediately renders three vertices from both the red and green sets unreachable, so we cannot proceed any further. This is the same kind of issue as with the earlier moduli-constructions - if the internal structure of the independent set is too "regular" (in the sense that all the gaps are the same length and the set itself is small) then this means there will be some shift of the independent set whose elements are hard to connect to v .

Let us illustrate the solution for $k = 7$ in its complete form.



It is clear that this construction generalises quite nicely to all odd k . Before we continue, however, note that until now we have shown that each vertex is in at least x independent sets of size x and y cliques of size y , but as to why there are not additional or larger independent sets or cliques, we have simply justified by visual consultation of the diagram. Of course, given the graph structure, one can formally prove this, but it is easier to use the following equivalent definition of (x, y) -graphs.

Definition: Minimal imperfect graphs II.

Given a graph $G = (V, E)$, we say the graph is an (x, y) -graph if it satisfies the following conditions:

- $\alpha(G) = x$, $\omega(G) = y$;
- $|V| = xy + 1$;
- For all vertices $v \in V$, $G - v$ can be decomposed into either x disjoint cliques of size y , or y disjoint independent sets of size x .

Bland, Huang and Trotter[1] proved this definition is equivalent to the presented before. In particular, it is clear why $\alpha(G - v)$ is x and $\omega(G - v)$ is y , as fewer colours is not possible by the size of the largest clique, and a larger clique is not possible

by the pigeonhole over the decomposition into independent sets. Moreover, this implies that $\alpha(G)$ is x and $\omega(G)$ is y , as if there was a clique or independent set of size 1 greater, it suffices to take v as a non-member of that set to obtain a contradiction.

More generally, for $k = 2n + 1$, one can do the following:

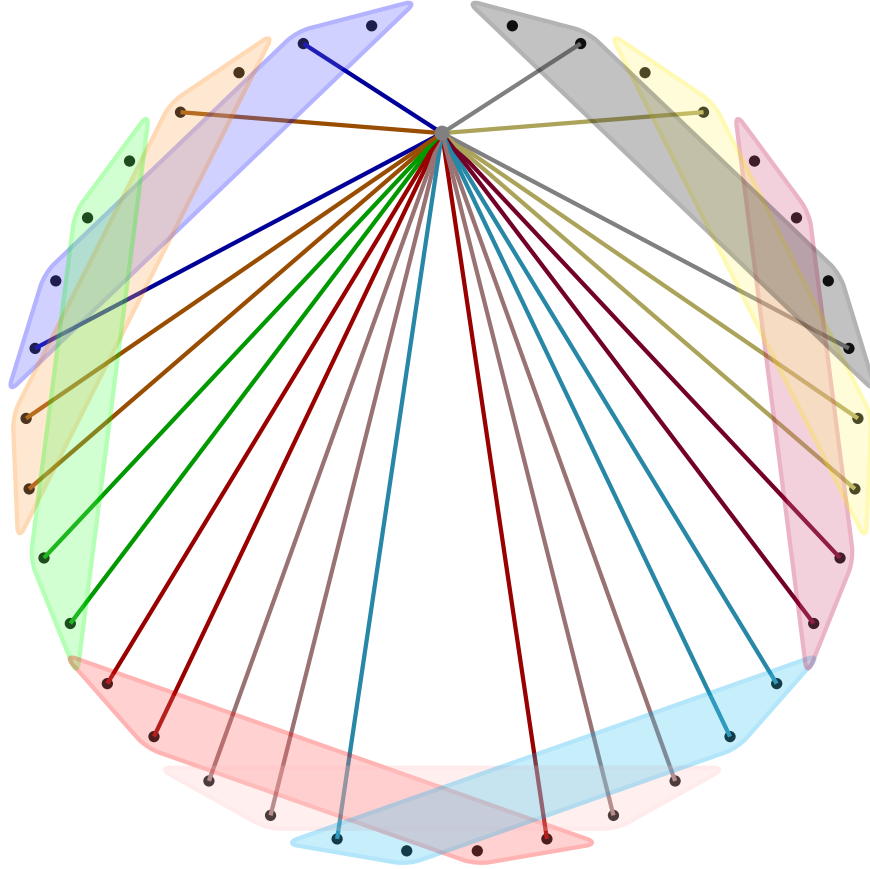
- Consider $\mathbb{Z}/(8n + 1)\mathbb{Z}$. We will construct a Cayley graph over this group.
- We want independent sets of the form $\{a, a + 1, a + 2n, a + 2n + 1\}$. In particular, removing 0, we can cover the rest by using $a \in \{1, 3, \dots, 2n - 1, 4n + 1, 4n + 3, \dots, 6n - 1\}$.
- This makes $\pm 1, \pm 2n - 1, \pm 2n$ or $\pm 2n + 1$ unusable for our generating set. We are forced to use 2 and $2n + 2$ in order to ensure there are two edges to the independent sets $\{1, 2, 2n + 1, 2n + 2\}$ and $\{6n - 1, 6n, 8n - 1, 8n\}$. Every other value is available to us, but we want to ensure cliques of size $2n$ that are identical when reflected. The easiest way to do this is to take the generating set $\{\pm 2, \pm 4, \dots, \pm 2n - 2, \pm 2n + 2, \pm 2n + 3, \dots, \pm 4n\}$. This contains at least two elements from each independent set. Note that rather than writing this generating set as an arithmetic progression with gap 2 along with an arithmetic progression with gap 1, it is actually more natural to write it as $\{\pm 2, \pm 4, \dots, \pm 2n - 2, \pm 2n + 2, \pm 2n + 4, \dots, \pm 6n - 2\}$, using the equivalence $\pm x = \mp(8n + 1 - x)$.
- This gives cliques of the form $\{a, a + 2, \dots, a + 2n - 2, a + 4n, a + 4n + 2, \dots, a + 6n - 2\}$. In particular, removing 0, we can cover the rest by using $a \in \{1, 2, 2n, 2n + 1\}$.
- By transitivity of the graph, we are done. Furthermore, it is clear that 0 is connected to at least two elements from each colour class (indeed, three elements in all but two of them).

Note the structure of the independent sets: n sets on the left and n on the right, each with two pairs of consecutive elements. It is very natural to now ask whether, instead of splitting the vertices into two groups like this, one might be able to split them into multiple groups. Indeed, this is clearly feasible. Let us now create a $(4, mn)$ -graph using the same methodology.

- Consider $\mathbb{Z}/(4mn + 1)\mathbb{Z}$. We will construct a Cayley graph over this group.
- We want independent sets of the form $\{a, a + 1, a + 2n, a + 2n + 1\}$. In particular, removing 0, we can cover the rest by using $a \in \{1, 3, \dots, 2n - 1, 4n + 1, 4n + 3, \dots, 6n - 1, 8n + 1, 8n + 3, \dots, (m - 1)n + 1, (m - 1)n + 3, \dots, (m - 1)n + 2n - 1\}$. Notice that we are just taking m different sequences of n odd integers with gaps between, rather than just two as we had before.

- This makes $\pm 1, \pm 2n - 1, \pm 2n$ or $\pm 2n + 1$ unusable for our generating set. We are forced to use 2 and $2n + 2$ for the same reasons as before. Every other value is available to us, but we want to ensure cliques of size mn that are identical when reflected. The easiest way to do this is to take the generating set $\{\pm 2, \pm 4, \dots, \pm 2n - 2, \pm 2n + 2, \pm 2n + 4, \dots, \pm 6n - 2, \pm 6n + 2, \dots, \pm 10n - 2, \pm 10n + 2, \pm 10n + 4, \dots, \pm (4m - 6)n - 2, \pm (4m - 6)n + 4, \dots, \pm (4m - 2)n - 2\}$. This contains at least two elements from each independent set. Note that whilst before we had all the even numbers up to $6n$ omitting $2n$ and $6n$, now we have all the even numbers up to $(4m - 2)n$ omitting $2n, 6n, 10n, \dots, (4m - 2)n$.
- This gives cliques of the form $\{a, a + 2, \dots, a + 2n - 2, a + 4n, a + 4n + 2, \dots, a + 6n - 2, a + 8n, a + 8n + 2, \dots, a + (m - 1)4n, a + (m - 1)n + 2, \dots, a + (m - 1)4n + 2n - 2\}$. In particular, removing 0, we can cover the rest by using $a \in \{1, 2, 2n, 2n + 1\}$. Notice the symmetry between the cliques and the representatives of the independent sets we picked earlier, and vice versa.
- By transitivity of the graph, we are done.

We illustrate the first example which was not covered by our previous even construction, $k = 10$, given by taking $m = 3, n = 3$.



We omit the internal edges of the graph as there are far too many and it is not useful to see them all represented. Via this construction, we can find graphs that are k -critical with no critical edges for all $(k - 1)$ nonprime.

2.3 Jensen's construction for all $k \geq 5$

There are two ways we would like to further generalise this family of constructions: for one, it would be desirable to find constructions for the case where $(k - 1)$ is prime. Additionally, it seems possible with some work to make each vertex connected to at least an arbitrary number of elements m of each colour class - meaning, if you remove less than m edges adjacent to any given vertex, the graph will remain k -chromatic.

In 1999, at the same time that Lattanzio published his result, Jensen[6] managed to find an alternative construction that satisfied the above for every $m > 1, k > 4$. Notably, the construction is quite different between k odd and k even. We will explain both constructions whilst showcasing four cases: $(k, m) \in \{(5, 2), (5, 4), (6, 2), (8, 6)\}$. As before, we will be constructing Cayley graphs on integer moduli groups.

Case 1: k is odd.

Jensen notes that in particular the case $m = 2$ yields an isomorphism of Brown's construction, so this is in some sense a generalisation of the prior result. However, this time, we do not create (x, y) -graphs but rather go through a laborious process in order to prove that there is a unique $(k - 1)$ -colouring of $G - v$.

We begin by taking the group we will be constructing a Cayley graph on. We take $\mathbb{Z}/N\mathbb{Z}, N := 2(k - 1)m\lceil m/2 \rceil + 1$. In the case $m = 2$, this gives $\mathbb{Z}/17\mathbb{Z}$. In the case $m = 4$, this gives $\mathbb{Z}/65\mathbb{Z}$.

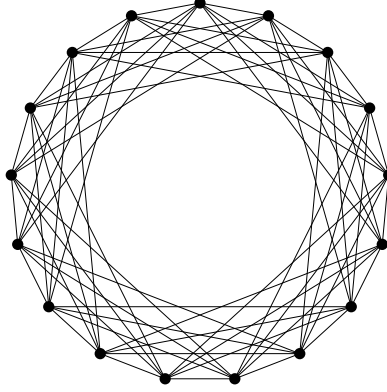
We now define the generating set. Note that $[a, b]$ in this context refers to the set of integers between a and b inclusive. We take the following set:

$$D_{m,k} := \{\pm 1, \pm 3, \pm 5, \dots, \pm 2m - 3\} \cup [\pm 2m - 1, \pm (k - 3)m + 1]$$

In the case $m = 2$, this gives $\{\pm 1, \pm 3, \pm 4, \pm 5\}$.

In the case $m = 4$, this gives $\{\pm 1, \pm 3, \pm 5, \pm 7, \pm 8, \pm 9\}$.

This gives the following graph for $m = 2$ (It is completely pointless to draw $m = 4$ due to the huge number of edges).



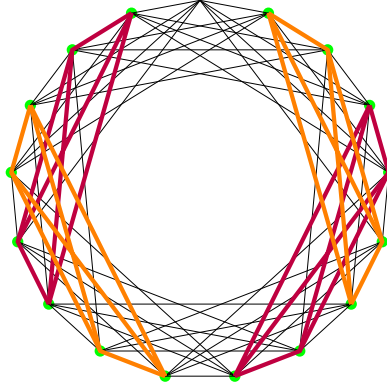
For the sake of clarity, we will therefore not draw all the edges in the graph for $m = 4$ in later diagrams, but rather only the pertinent ones.

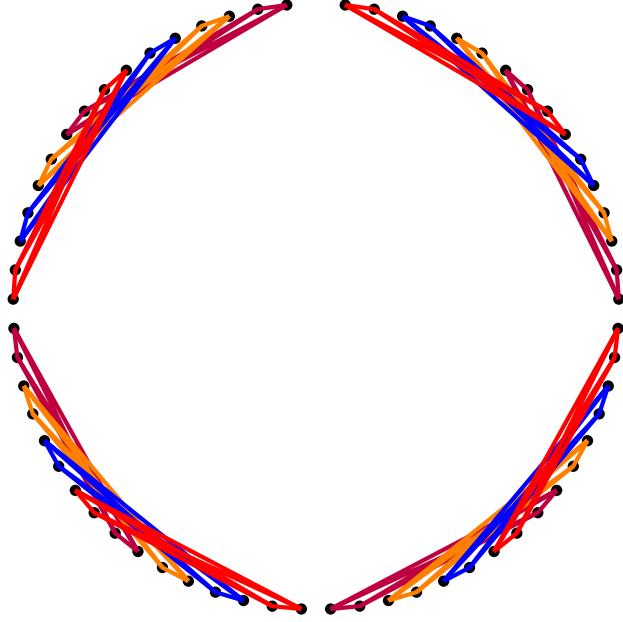
Consider now the induced subgraph G_i of $G - 0$ given by the vertex set $\{i, i + 1, \dots, i + (k - 1)m - 1\}$, for $i \in [1, N - (k - 1)m + 1]$. We now endeavour to prove structural qualities of H_i ; therefore, for the following, we may assume by transitivity $i = 1$.

The next step will be proving that $\chi(G_1) \geq k - 1$. In order to do this, we will just show that the vertex set of G_i can be partitioned into m disjoint cliques. This will also imply $\alpha(G - 0) \leq m$. This is not difficult: note that we can find cliques of the form

$$\{a, a + 1, a + 2m, a + 2m + 1, \dots, a + (k - 3)m, a + (k - 3)m + 1\}$$

containing $(k - 1)$ elements each, taking $a \in \{1, 3, \dots, 2m - 1\}$. A colour or independent set can contain at most one element from each.





Before proving the existence of a colouring on $G - 0$, let us first prove unicity. We will start by proving any valid $(k - 1)$ -colouring of $G - 0$ is periodic, with period $m(k - 1)$ (in the sense that if ϕ is a colouring, $\phi(x) = \phi(x + m(k - 1))$), **not** allowing for indices to pass over 0 and loop around). But this is trivial - if such a colouring exists, then for each H_i there are exactly m elements of each colour, meaning shifting i forward by 1, we must replace the outgoing vertex by one of the same colour.

Now consider the symmetric relation $\sim$ defined by $a \sim b$ if, for all valid $(k - 1)$ -colourings ϕ of $G - 0$, $\phi(a) \neq \phi(b)$. We want to add as many elements to the set $A := \{b \mid a \sim b\}$ as possible, to make the colouring forced. Note that this relation is clearly neither reflexive nor transitive. By definition, if $|a - b| \in D_{m,k}$ then $a \sim b$. By the aforementioned periodicity, it follows that if $|a - b| \bmod n(k - 1) \in D_{m,k}$ then we can draw the same conclusion. We also want to account for the restrictions imposed by the edges crossing over 0. In particular, if a is in G_1 and b is in $G_{N-n(k-1)+1}$ then, because of the one extra element in between them (0) there is an edge from a to b if and only if $|b - a| - 1 \bmod n(k - 1) \in D_{m,k}$. By periodicity, this means if $|b - a| - 1 \bmod n(k - 1) \in D_{m,k}$ then $a \sim b$. Notably, being connected to 0 should be disregarded when expanding the set A .

For example, consider the diagram for $m = 2$ above: if vertices are numbered anticlockwise and 0 is at the top, then clearly by the first remark 1 has a different colour to 2, 4, 5 and 6 by virtue of direct connection, and 10, 12, 13 and 14 by periodicity. Furthermore, 9 is connected to 10, 12, 13 and 14 (which does not add any information) but also 8, 6, 5 and 4, meaning by periodicity 1 cannot share a colour with 8 either. Finally, looking at the edges crossing over 0, 1 is also a different colour to 0, 15, 14 and 13 by direct connection, and 7 by periodicity. Having a different colour to 0 means nothing here, however, because we are considering a colouring of $G - 0$ anyway. This means, of the elements of G_1 , 1 can only share a colour with 3.

Due to periodicity we can therefore restrict ourselves to finding a valid colouring for G_1 . Clearly, we will have to assign one colour from each clique to each of the colours, if we want to only use $(k - 1)$ colours. Moreover, the set of valid differences between a and another element of its independent set by the above argument is $(A - a)^c \cap [a - 1, n(k - 1) - a]$, giving

$$\{2, 4, \dots, 2m - 2\} \quad \forall a.$$

Here in particular is the reason $(k - 1) > 4$ is necessary, as otherwise we would not get this set specifically.

Recall that we just showed that this is indeed the case for $m = 2 - 1$ can only share a colour with $1 + 2 = 3$, meaning for $m = 2$ the $(k - 1)$ -colouring of $G - 0$ is forced. Likewise, if you do the calculations for the case $m = 4$, you will see that 2, 4 and 6 are the only valid differences between two elements of an independent set in G_1 .

We first claim that the only possible colouring is by using the independent sets

$$I_1 = \{1, 3, \dots, 2m - 1\}$$

$$I_2 = \{2, 4, \dots, 2m\}$$

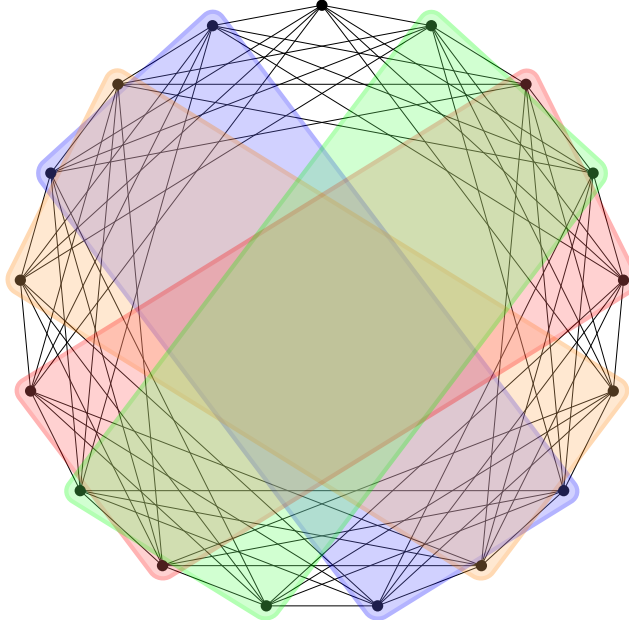
$$I_3 = \{2m + 1, 2m + 3, \dots, 4m - 1\}$$

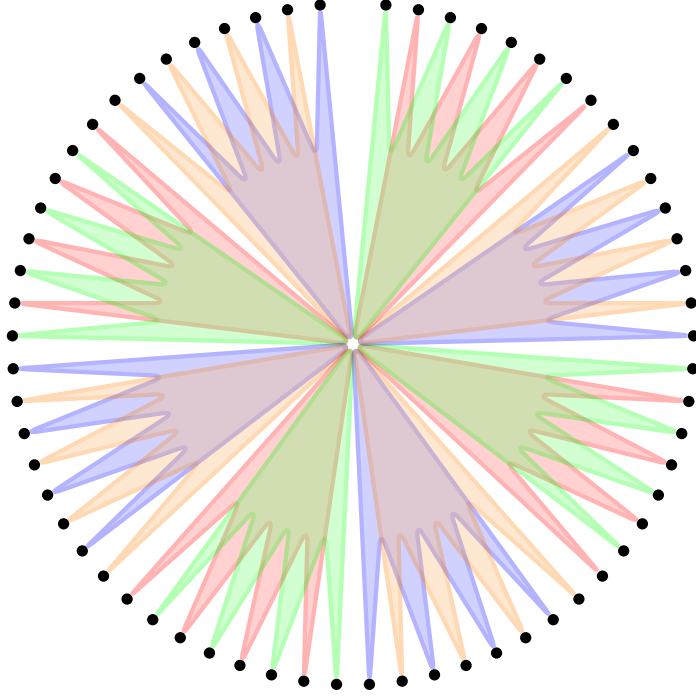
$$I_4 = \{2m + 2, 2m + 4, \dots, 4m\}$$

$\vdots$

$$I_{k-2} = \{(k - 3)m + 1, (k - 3)m + 3, \dots, (k - 1)m - 1\}$$

$$I_{k-1} = \{(k - 3)m + 2, (k - 3)m + 4, \dots, (k - 1)m\}.$$





Henceforth, in our diagrams for large graphs we will no longer draw convex polygons around the different vertices in the colour classes as the polygons would be indistinguishable. We will draw stars instead.

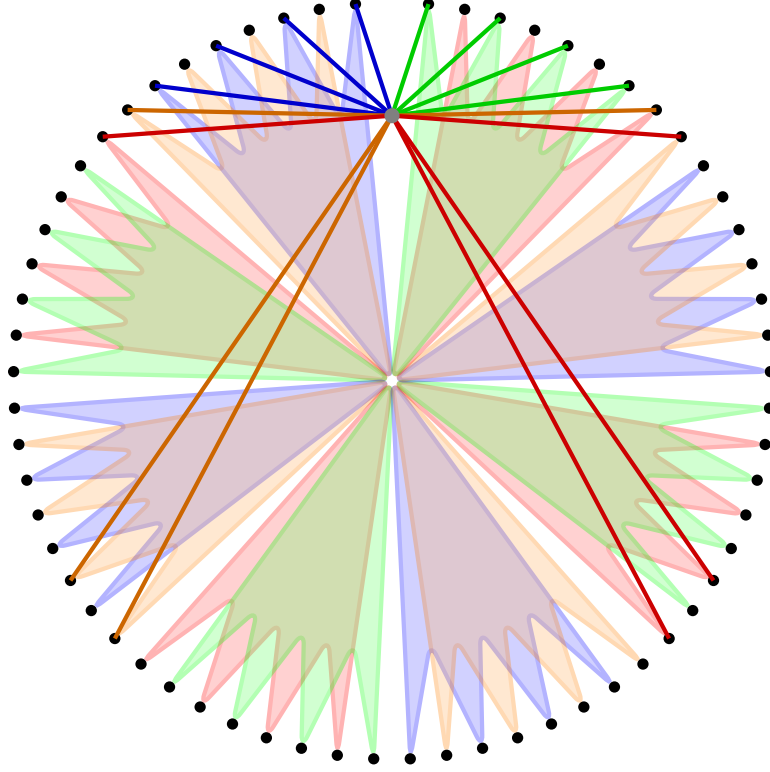
This is clearly valid as one can check that not only does this work for G_1 but the extension via periodicity to $G - 0$ also remains independent. It remains to prove unicity. This is not difficult: in order to use only $(k - 1)$ colours, we must have exactly m elements in each colour, meaning that given that A_1 has 1, the rest of the elements fall into place (since the set of valid differences is of size exactly $m - 1$). Similarly for $2 \in A_2$, and each time we go to the smallest element not yet assigned and find that the rest of the independent set follows naturally.

It remains to show that we can now ensure that the number of edges between 0 and any colour class is at least m . To do so, we will add additional elements to the generating set $D_{m,k}$ that do not increase $\chi(G - 0)$. It is easy to see that the number of edges from 0 to the first colour I_1 and the last colour I_{k-1} is already m (for example, for $m = 4$, the edges $\{\pm 1, \pm 3, \pm 5, \pm 7\}$ suffice). Additionally, the large interval in the generating set $[\pm 2m - 1, \pm (k - 3)m + 1]$ ensures 0 is connected at least m times to $I_3, \dots, I_{k-3}$. However, the only elements of I_2 and I_{k-2} that 0 is connected to are $\{2m - 1, N - (k - 3)m - 1\}$ and $\{N - 2m + 1, (k - 3)m + 1\}$ respectively. We therefore need to add $2m - 4$ elements to the generating set. It will clearly suffice to add $m - 2$ edges to A_2 as by symmetry A_{k-2} will also gain $m - 2$ edges if we add additive inverses. Let r be an integer such that $r \geq \lceil (m - 1)/2 \rceil$. We add

$$\bigcup_{1 \leq i \leq r} \{\pm i(k - 1)m + 2m - 1, \pm i(k - 1)m + (k - 3)m + 1\}$$

to $D_{m,k}$. By our choice of N , these edges are new and distinct from each other as well as the existing edges. It is also easy to verify that the introduction of these edges does not cause any problems for our coloration of $G - 0$. For instance, for $m = 4$ we would add ± 23 and ± 25 , giving us a final generating set of

$$\{\pm 1, \pm 3, \pm 5, \pm 7, \pm 8, \pm 9, \pm 23, \pm 25\}.$$



Case 2: k is even.

We begin by taking the group we will be constructing a Cayley graph on. We take $\mathbb{Z}/N\mathbb{Z}, N := 2(k-1)m[(m+1)/2] + 1$. In the case $k = 8, m = 2$, this gives $\mathbb{Z}/57\mathbb{Z}$. In the case $k = 6, m = 5$, this gives $\mathbb{Z}/151\mathbb{Z}$. Henceforth, we will only use the value of m to distinguish the two cases. The choices of k and m we will be following might seem arbitrary, but for instructive purposes it is useful to have one case in which $m > 4$ and one in which $k > 6$.

We now define the generating set. Note that $[a, b]$ in this context refers to the set of integers between a and b inclusive. We take the following set:

$$D_{m,k} := \{\pm 1, \pm 3, \pm 5, \dots, \pm 2m-3\} \cup [\pm 2m-1, \pm (k-4)m+2] \cup [\pm (k+2)m-1, \pm 2(k-2)m+1]$$

In the case $m = 2$, this gives $\{\pm 1, \pm 3, \dots, \pm 8, \pm 19, \dots, \pm 25\}$.

In the case $m = 5$, this gives $\{\pm 1, \pm 3, \pm 5, \pm 7, \pm 9, \dots, \pm 12, \pm 39, \dots, \pm 41\}$. Both of these graphs have far too many edges for an illustration of the full graph to offer any instructive purpose.

Consider now the induced subgraph G_i of $G - 0$ given by the vertex set $\{i, i + 1, \dots, i + 2(k - 1)m - 1\}$, for $i \in [1, N - 2(k - 1)m + 1]$. Similarly, we endeavour to prove structural qualities of G_i ; assuming by transitivity $i = 1$. Note that in the even case, we take subgraphs twice the size of the odd case.

Now we will show the maximal size of an independent set in G_1 is $2m$. However, we will proceed differently to before: this time, we will consider the possible elements of the independent set U containing 1. We will then write U as the union of $m - 1$ (not necessarily disjoint) sets, such that each of these sets U_i has size 10 and, letting $U_i = u_1, \dots, u_{10}$ in increasing order, we have in particular that the five subsets

$$\{u_1, \dots, u_4\}; \{u_3, \dots, u_6\}; \{u_5, \dots, u_8\}; \{u_7, \dots, u_{10}\}; \{u_9, u_{10}, u_1, u_2\}$$

are all cliques. This will prove in particular an independent set contains at most 2 elements from each U_i as it is equivalent to taking vertices from a pentagon without taking two adjacent vertices; one can take at most 2.

First consider the structure of U ; it must be a subset of

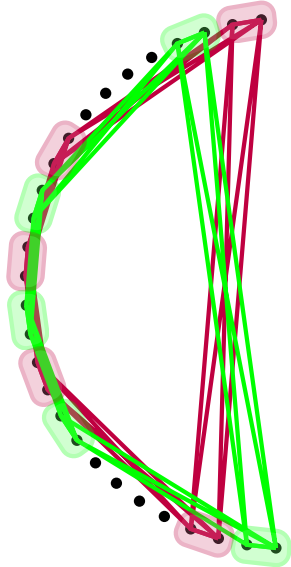
$$[1, 2m] \cup [(k - 4)m + 1, (k + 2)m] \cup [2(k - 2)m + 1, 2(k - 1)m]$$

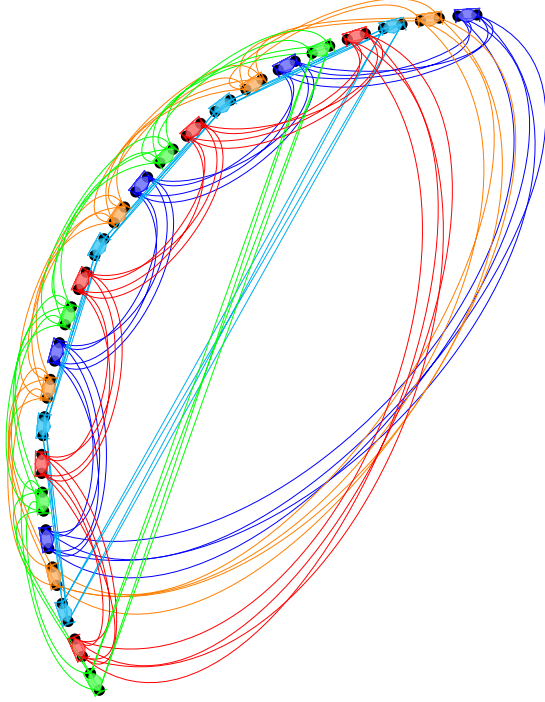
and one can easily check the other elements of G_1 are blocked by the chosen generating set.

Now just take U_i as (letting $j = i + 1$ so the equation fits on one line!)

$$\{i, j, (k - 4)m + i, (k - 4)m + j, (k - 2)m + i, (k - 2)m + j, km + i, km + j, 2(k - 2)m + i, 2(k - 2)m + j\}$$

and we see that $U_1, \dots, U_m$ covers U and each U_i has the described structure.





Note that we are just seeking to cover elements which could be in the independent set containing 1 and not all of G_1 , hence why in the case $m = 2$ we do not cover all the vertices.

We have therefore proved that $\alpha(G_1) \leq 2m$.

The next section of the proof proceeds identically to the odd case - establishing a set of vertices A which must have a different colour to a . The only difference is, given that periodicity is now $\text{mod } 2(k-1)m$, and the generating set is different, we now obtain that the elements of G_1 that can share a colour with a (those of the form $(A - a)^c \cap [a - 1, n(k-1) - a]$) are now the set:

$$\{\pm 2, \pm 4, \dots, \pm 2m - 2\} \cup [\pm(k-4)m + 3, \pm(k+2)m + 2] \quad \forall a.$$

As this set does not have cardinality $2m - 1$, the proof of unicity is now no longer instant.

We must embark on a different approach. Our approach is motivated by the structure of the independent sets. Notably, if we study the generating set, we can easily spot the intended unique good $(k-1)$ -colouring of $G-0$. We can partition it into $k-1$ independent sets, which take the form

$$I_j = \{(j-1)m+1, (j-1)m+3, \dots, (j+1)m-1, (j+k-3)m+2, (j+k-3)m+4, \dots, (j+k-1)m\}$$

for odd j and

$$I_j = \{(j-2)m+2, (j-2)m+4, \dots, jm, (j+k-3)m+2, (j+k-2)m+1, \dots, (j+k)m-1\}$$

for even j . Notably, we once again have two groups of m vertices which are each just arithmetic progressions with gap 2, so we will endeavour to prove that this structure is necessary. In particular, we see that the set above cannot have too many “large gaps” (greater than $2m-2$ and hence at least $(k-4)m+3$) without their combined length being larger than the largest valid gap $((k+2)m+2)$. We will first prove that there is at most one large gap, and then analyse the necessary structure of the two groups on either side of this gap.

Let A be an arbitrary independent set in G_1 , and a_1 be its smallest element. We now recursively define b_i as the number of elements of A between a_1 and $a_1 + 2m - 2$ inclusive (so at least 1, since a_1 is always in this set) and a_{i+1} the smallest element of A greater than $a_i + 2m - 1$. If at any point there are no elements greater than $a_i + 2m - 1$ then set all subsequent a_{i+k} (and the corresponding b_{i+k}) to be equal to $+\infty$. Note that each a_i is the first element of a “group” of vertices with pairwise small differences, and b_i is just the size of each group. We would like to show that $a_3 = \infty$, implying only two groups exist.

Since the smallest valid gap between two elements of an independent set is 2, and conditional of the gap being greater than $2m-2$ is $(k-4)m+3$, we know that $a_{i+1} \geq a_i + (k-4)m+1+2b_i$, since every element in the set giving b_i (aside a_i itself) extends this gap by at least 2. Applying this inequality $l-1$ times, we get $a_{l+1} \geq a_1 + (l(k-4)m+1) + 2 \sum_{i=1}^l b_i$.

Now take the largest $l \geq 0$ such that $a_{l+1} < \infty$ (this must exist because $a_1 < \infty$ and $|A| \leq 2m \implies a_{2m+1} = \infty$). We have that

$$a_{l+1} + 2b_{l+1} - 2 \leq \max(A) \leq a_1 + (k+2)m - 2$$

where the lower bound on the largest element of A is due to the gaps being at least size 2 from a_{l+1} upwards and the upper bound is from the largest valid gap given above. If we combine this with the lower bound on a_{l+1} , then clearly $l < 2$, because otherwise, as $k > 6$, two or more of the large gaps (of size at least $l(k-4)m$) along with the small gaps is larger than the largest possible gap. Note that we must implicitly use here that the total number of gaps is at least $2m-1$, as every colour must have exactly $2m$ elements if we can only use $(k-1)$ colours. Consider also that we clearly cannot have only one group, as the largest gap is at least $2(2m-1)$ and so not every pair of elements in A can have a difference less than $2m-1$.

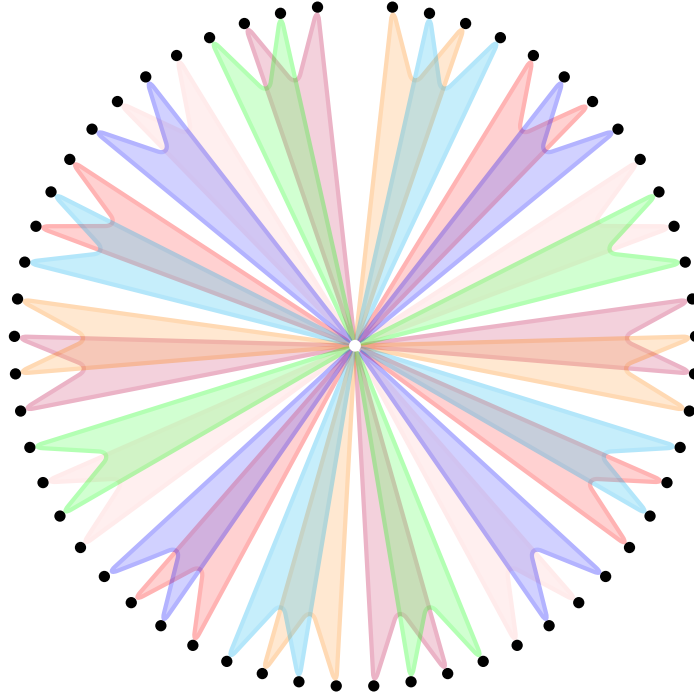
Knowing that we have two groups, we can therefore write A as the union of A_1 and A_2 , where A_1 is composed of elements at distance at most $2m-2$ from its smallest element

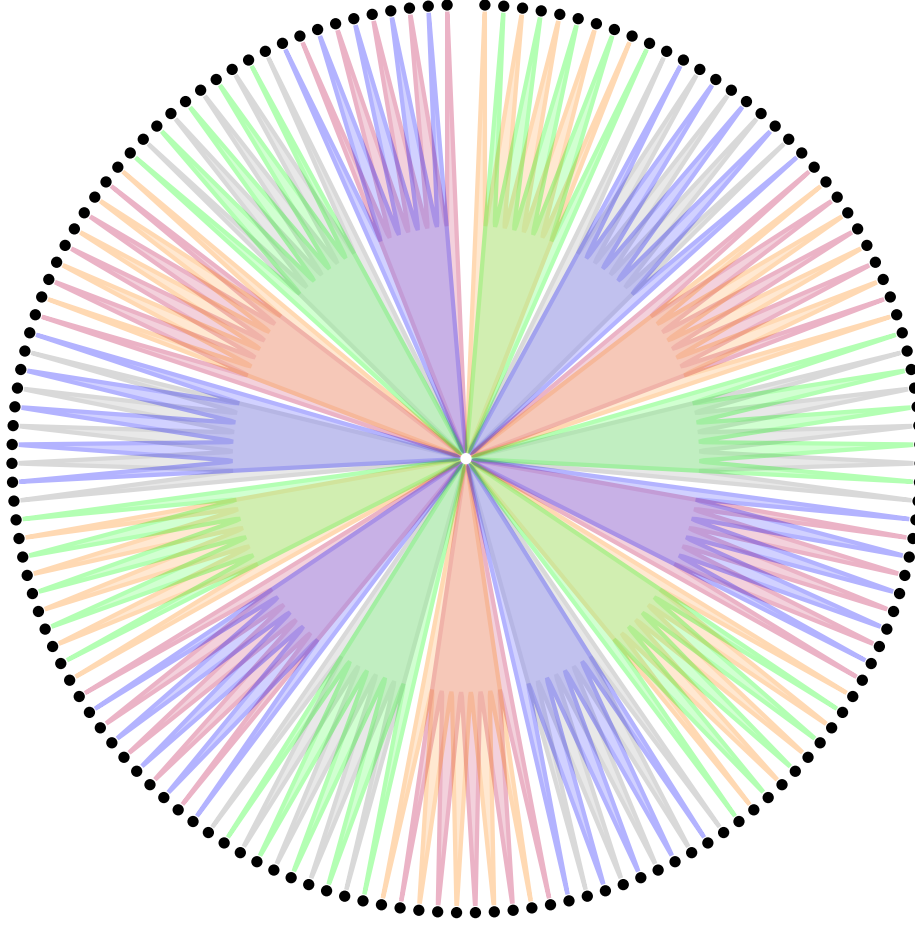
a_1 , the same for A_2 and its minimum a_2 , and a_2 is at least $a_1 + (k - 4)m + 1 + 2|A_1|$. Furthermore, it is clear that each group must contain m elements as otherwise the larger one of the two will have a largest gap greater than $2(m - 1)$, contradicting its definition. In order for all the inequalities to be tight, the gaps between consecutive terms in each group must be exactly 2 and so the structure of A is

$$\{a_1, a_1 + 2, \dots, a_1 + 2m - 2, a_2, a_2 + 2, \dots, a_2 + 2m - 2\}.$$

What are the valid values for $|a_1 - a_2|$? Clearly, the set of valid gaps must contain $|a_1 - a_2|$, as well as $|a_1 - a_2| - 2, |a_1 - a_2| - 4, \dots, |a_1 - a_2| - 2m + 2$ and also $|a_1 - a_2| + 2, |a_1 - a_2| + 4, \dots, |a_1 - a_2| + 2m - 2$. Since all of these must lie in the set of large gaps, it follows that $|a_1 - a_2| \in [\pm(k - 2)m + 1, \pm km]$.

Now, we can cut up $[1, 2km]$ into groups of the form $\{a_1, a_1 + 2, \dots, a_1 + 2m - 2\}$ and assign two to each colour under the restriction above. It is clear that there is only one way to cut it up - the same argument as in the odd case, where we take the smallest element not yet chosen and create a group starting with it works. Now, applying the lower bound on the difference between a_1 and a_2 , we see that the first $k - 2$ groups (ordered by minimal element) must belong to different colours. Considering the remaining groups, it is easy to see that we are forced to pick the penultimate one for the remaining colour, since $(k - 2)m + 1$ belongs to this group, has not yet been assigned a colour and is now incompatible with the other colours. Now $(k - 2)m + 2$ is now only compatible with the first colour, and we can continue making deductions in this manner to obtain the unique colouring we noticed before.





Now the final question remaining to us is how many times 0 is connected to each vertex. Similarly to before, the number of edges from 0 to the first colour I_1 and the last colour I_{k-1} is already m due to the large set of small odd edges (for example, for $m = 5$, the edges $\{\pm 1, \pm 3, \pm 5, \pm 7, \pm 9\}$ suffice). Furthermore, the two large intervals in the generating set $[\pm 2m - 1, \pm(k - 4)m + 1] \cup [\pm(k + 2)m - 1, \pm 2(k - 2)m + 1]$ ensures 0 is connected at least m times to $I_3, \dots, I_{k-3}$ **but only if** $k > 6$. Jensen, in particular, does not note this edge case in his paper; if $k = 6$ then one only gets 4 edges to I_3 . It remains again to add more edges to address I_2 and I_{k-2} - this time, we note that they already have 4 edges each. If $m > 4$, we proceed nearly identically to before - take $r = \lfloor m/4 \rfloor$ and just add to the generating set

$$\bigcup_{1 \leq i \leq r} \{\pm j + 2m - 1, \pm j + (k - 4)m + 2, \pm j + (k - 2)m + 1, \pm j + 2(k - 2)m + 2\} \quad \text{where } j := i2(k - 1)m.$$

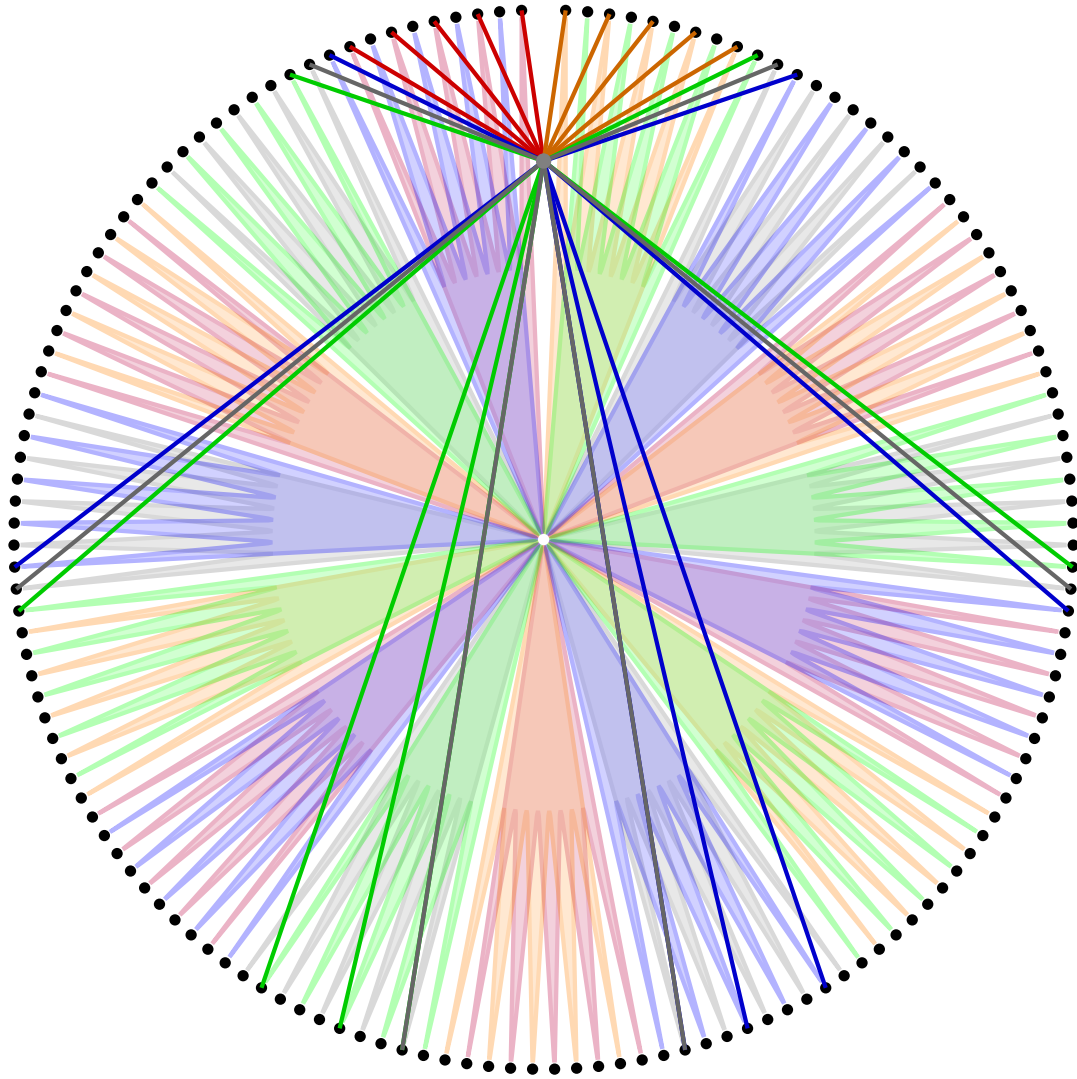
and we can easily verify this adds at least $m - 4$ edges to I_2 and I_{k+2} when $k > 6$ without introducing any issues. Most importantly, in the specific case $k = 6$, this introduces edges to I_3 as well (rather than introducing 4 edges to each of I_2 and I_4 , each value of i in the union adds 3 each to I_2 and I_4 , and 2 to I_3 , with the edges of the form

$\pm i2(k-1)m + 2(k-2)m + 2$ going to I_3). This allows the case $k = 6$ in Jensen's proof to be saved by simply replacing m with $n = \lceil 4m/3 \rceil$ from the start.

For instance, for $m = 5$ we would add $\pm 59, \pm 62, \pm 66$ and ± 82 , giving us a final generating set of

$$\{\pm 1, \pm 3, \pm 5, \pm 7, \pm 9, \pm 10, \pm 11, \pm 12, \pm 39, \pm 40, \pm 41, \pm 59, \pm 62, \pm 66, \pm 82\}$$

and as we can see in the diagram, of the 8 additional edges, 3 go to A_2 or A_4 and 2 go to A_3 . This clearly holds for higher values of m because colours are modulo invariant $2(k-1)m$.



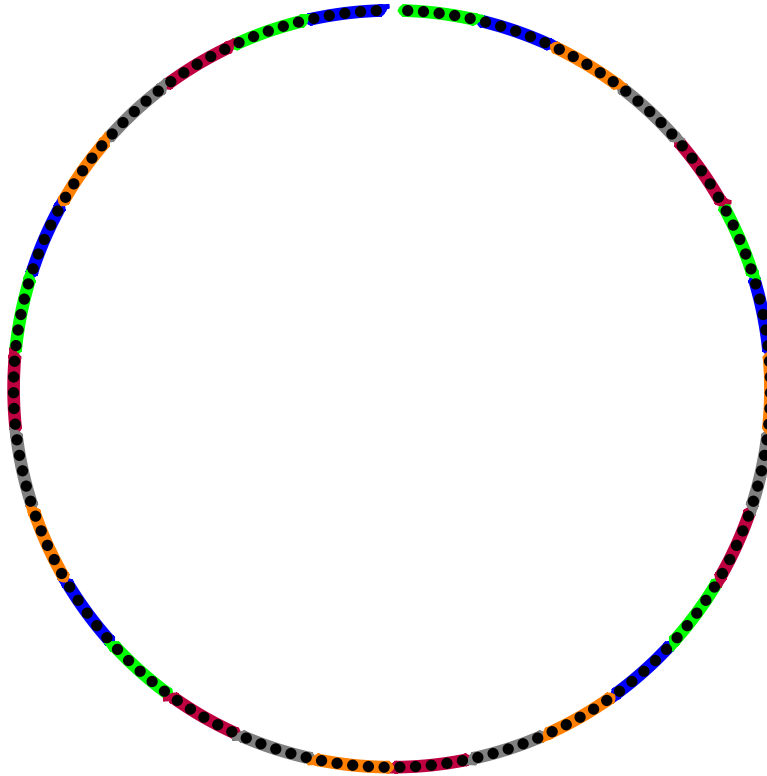
3 Erdős' conjecture and the future

We'll continue tomorrow — if I live.

Paul Erdős

3.1 The final frontier: $k = 4$

What is the motivation behind Jensen's construction for even k , and why can it not be modified to work for $k = 4$? As we saw for Lattanzio's construction, which is after all a very natural generalisation of Brown's original construction, taking independent sets that are the same up to rotation simply does not work when there are an odd number of independent sets. Jensen's solution was simply to allow for different structures of independent sets, albeit extremely similar in nature. Let us do something crafty, and take the coloration of $\mathbb{Z}/151\mathbb{Z}$ on the previous page and apply to it the function $f(x) = x/2$, which is a group isomorphism as $\gcd(151, 2) = 1$.



This isomorphism has the nice effect of turning groups into consecutive sequences. We can see that the distance between two groups (as a multiple of m) is either $0, 2$ or $3 \pmod{5}$,

albeit in the third case with an extra vertex between the two groups. More generally, for arbitrary k , it would be 0 or $(k-1)/2 \pmod{k-1}$. This forbids edges in the ranges $[0, 4] \cup [21, 24] \pmod{25}$ - but only up to 75 , since at most three groups in a row are separated by multiples of 5 - and $[7, 19] \pmod{25}$ respectively for $k = 5$, for instance. Jensen is now free to add all other edges modulo 25 , and his true stroke of genius is finding the proof that (a subset of) these edges (is) are enough to guarantee uniqueness of the colouring. This fact is extremely intuitive due to the large number of edges one could potentially add, but proving the bound on the maximum size of independent sets is very difficult, and necessitates ingenuity. The remainder of the proof is quite natural but still remarkable nonetheless.

It is worthwhile noting the reasons for why Jensen is able to improve from 1 to k edges in the odd case - as we stated before, in Lattanzio's construction, several of the colours have 3 inbound edges, but two have 2 . By applying the same isomorphism to Jensen's even construction, we note that it is extremely similar; it has symmetric independent sets that are all identical, and the main change is just taking larger consecutive groups than just 2 at a time. Of course, the main issue with this is that we no longer have (x, y) -graphs and cannot use the beautiful result noted before; but it still seems likely that the colouring is unique and this is what Jensen set out to prove (and successfully managed to do).

Why, then, does the construction not work for $k = 4$? As noted at the end of the previous section, it already comes dangerously close to failing for $k = 6$ (and Jensen may have overlooked this). $k = 3$ just exacerbates the problem - I_2 just has no way of getting more edges. Indeed, unlike every other case, it does not seem like a given that Dirac's conjecture is true, and it might be very possible that the converse is true.

The main issue for this case is that all the constructions we have seen so far have been *circulant*, a term used in literature to refer to Cayley graphs on a finite cyclic group. This is quite useful because the transitivity is crucial, but even if we proved all 4 -critical circulant graphs have a critical edge, it would not even prove that there are no transitive constructions possible, let alone prove the result for all graphs. Note that even the circulant case is hard to answer, as all our constructions have relied on the uniqueness of the colouring of $G - 0$ and we cannot assume this is always the case for 4 -critical graphs. The difficulty in this case arises from the fact that it is very hard to qualify or quantify what it means to be 4 -critical.

3.2 On the problem of removing multiple edges

Unfortunately, from Jensen's construction we cannot remove *any* set of $m - 1$ edges and assume this does not reduce the chromaticity. However, it does seem like being able to remove $m - 1$ edges at the same vertex is quite strong, since if you wanted to reduce the chromaticity, this seems like the fastest way to do it. Jensen did, however, check small cases and found that this property held for many of his constructions. He conjectured that infinitely many choices of N would allow you to remove any set of $m - 1$ edges³. This would settle Erdős' conjecture in the affirmative.

³In our explanation, we gave N as a fixed value; but it is clear from the periodicity modulo $2m(k-1)$ that we can add multiples of $2m(k-1)$ to N whilst retaining all the same properties.

The main difficulty in proving this lies in the fact that this is equivalent to the statement that, for any k -good colouring of G , there are m edges between any vertex and all elements of other colour classes. However, there are many, many k -good colourings of G that we have not considered yet - we took a $(k - 1)$ -good colourings of $G - v$ and added a vertex, but we could also have (k) -good colourings of $G - v$ and then add a vertex, and we have no good information on what an arbitrary k -good colouring of G might entail.

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